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Cluj-Napoca, Romania | Portfolio

July 15, 2026

Hiring Team

Bertrandt

Cluj-Napoca, Romania

Application for Model Based Developer (Hybrid)

Dear Bertrandt Hiring Team,

I am applying for the Model Based Developer position in Cluj-Napoca. I recently graduated in Automation and Computer Science from the Technical University of Cluj-Napoca with a final average of 9.25/10. This role caught my attention because it sits exactly where I want to build my career: turning requirements and control logic into models that can be tested, reviewed, and eventually run on embedded hardware.

My first professional experience in model-based development was at Bosch, from July 2024 to January 2025. I worked on a six-component electric-vehicle drivetrain model using Simulink and ASCET, supported validation activities in TPT, generated C code, and deployed software on an STM32 microcontroller. It was a short placement, but it confirmed something important: this is the kind of engineering work I want to keep doing. I enjoy the patient, methodical work behind a reliable model - understanding what the system should do, tracing unexpected behavior, checking assumptions, and iterating until the results make sense.

My bachelor's thesis strengthened the mathematical side of that interest. I worked on nonlinear control and feedback linearization, translating control theory and academic literature into a practical MATLAB implementation. That process taught me to be comfortable moving between equations, software, simulation results, and technical explanations - a combination I would like to keep developing in an automotive environment.

I already have a useful foundation for several parts of the role, especially MATLAB, Simulink, code generation, and MiL testing with TPT. Stateflow, AUTOSAR, MISRA, and Polyspace are natural next steps for me, and I am genuinely interested in learning them properly within a team that combines mentoring with real project responsibility. I tend to work in a structured way, document what I do, and ask questions early when requirements are unclear. I also enjoy working with other engineers and discussing the reasoning behind a technical decision, not only its final implementation.

Bertrandt's work in e-mobility and embedded systems is a strong match for the work I enjoyed at Bosch and the control engineering path I chose at university. I am based in Cluj-Napoca, work fluently in English, and have an A2 level in German that I intend to improve. I would be glad to bring my current experience, curiosity, and commitment to your team while growing into the responsibilities of the position.

Thank you for considering my application. I would welcome the opportunity to discuss how my background and motivation could contribute to Bertrandt's model-based development projects.

Kind regards,

Carla - Maria Băraștean